LIGHTYEAR

MS 96 Isolated agent verification and operator walkthroughs

Reconciled October 14 2026.

Lightyear Verify gives an agent a controlled way to submit a candidate and inspect its result while keeping evaluation inputs and signing authority in a separate judge. This milestone brings the toolkit, confidential evaluation controls and operator walkthrough records into the searchable library. It demonstrates the declared public-fixture workflow, not autonomous development effectiveness.

Delivered capability

The stdio MCP toolkit reads approved public material and submits Java JARs to an operator-owned Linux judge. The judge executes them under a different OS identity with private inputs, no network and bounded output files. UUID idempotency, durable attempt reservations and cumulative budgets survive restarts. Interrupted accepted attempts do not silently rerun or regain their budget.

The default documented budget is five submissions and 25 reserved execution minutes. Signed receipts, a chained journal and offline replay retain verdicts and refusal evidence. Confidential evaluation uses closed diagnostics; normalization remains a human-reviewed proposal rather than an agent-controlled rule change.

Evidence and remaining work

The original acceptance exercised a good public INTCALC artifact, rounding and omission faults, timestamp divergence, repeated-diagnostic alerts, exhausted budgets and offline replay. Security checks exercised separate users, protected files, candidate process cleanup and receipt tampering.

Howard's retained Apple Silicon and Ubuntu arm64 records report the manual walkthrough and Claude Code and Kiro public-fixture results. Those are operator-reported observations, not independent attestation. The October 9 graph-off acceptance is separate from the later automated 15-step walkthrough, which was tested offline but had no completed Linux VM execution in its publication record. Codex live-client completion and x86_64 acceptance are not established by those records.

Real customer equivalence, hosted multi-tenancy and general noninterference are outside this claim. A trusted terminal journal head is needed to detect removal of a valid suffix. New client runs and model use require their own authorization.

Supporting records

- Toolkit and isolation acceptance
- Confidential evaluation and durable budget hardening
- Smoke kit and dated client results
- Graph-off acceptance and automated walkthrough